NEREID / 532

SUBSEA OPTICAL METROLOGY / CONCEPT ENGINEERING

SENSOR DATA & VERIFICATION BASIS

Separate data sheet / assumed design targets / derived quantities / validation plan

01 SENSOR SPECIFICATION / ALL VALUES ARE CONCEPT TARGETS

PARAMETER	TARGET / DESIGN ASSUMPTION
Principle	Pulsed direct time of flight; bistatic receiver
Wavelength / pulse width	532 nm / 2 ns nominal
Pulse energy / rate	5 uJ / 20 kHz nominal
Optical average power	0.10 W, derived before optical losses
Range objective	1-15 m in clear water; not demonstrated
Precision objective	<=20 mm, 1 sigma at 5 m under test conditions
Absolute accuracy	Unspecified until calibration and tank test
Scan field / angular step	30 deg in water / nominal 0.03 deg
Scan throughput	20,000 shots/s; <=20 lines/s at 1,000 shots/line
Receive aperture / FOV	D60 clear window / >=40 deg full optical FOV
Receiver / digitizer	Silicon APD + TIA / 2 GS/s target
Power / electrical	18-30 VDC; 35 W nominal / 60 W peak budget
Data / timing	100BASE-TX / PPS / PTP support target
Depth / temperature	300 m target / water 0 to +35 C target
Mechanical envelope	D160 max x 359 incl. connector; 320 body
Mass / buoyancy	TBD by CAD mass model; no claimed value
Output	XYZ, intensity, time, scan angle, quality flags

02 CALCULATIONS / NOT MEASURED SENSOR DATA

A TIME OF FLIGHT / n_group = 1.34 assumed
c0 = 299,792,458 m/s; v_group ~ 223.73 x 10^6 m/s
At 5 m: dt ~ 44.70 ns (monostatic equivalent)
20 mm range change -> ~179 ps timing change

B SAMPLING / SCAN GEOMETRY
30 deg / 1,000 samples ~ 0.03 deg per shot
At 5 m: angular pitch ~ 2.62 mm near boresight
Swath at 5 m: 2R tan(15 deg) ~ 2.68 m
Beam divergence target 2 mrad -> ~10 mm growth at 5 m
Sample pitch does not equal spatial resolution.

C EXTERNAL PRESSURE / seawater density 1,025 kg/m^3
p_gauge = rho*g*h ~ 3.02 MPa at 300 m (g=9.81)
On D60 clear aperture: F ~ p*A = 8.53 kN
Window seat load also depends on unsupported diameter.
No collapse, fatigue, seal or window stress analysis done.

D ATTENUATION EXAMPLE / homogeneous water only
Two-way transmission factor ~ exp(-2*c_att*R)
At 10 m: c_att=0.10/m -> 0.135; 0.30/m -> 0.00248
Excludes geometric loss, reflectance and receiver noise.
No link-budget closure or range guarantee is implied.

03 OUTPUT RECORD / PROPOSED WIRE FORMAT

FIELD	TYPE / UNIT	MEANING
timestamp_ns	uint64 / ns	TAI epoch; validity flag required
xyz_sensor	3 x float32 / m	Right-handed sensor coordinate frame
range_tx	float32 / m	Distance along calibrated transmitted ray
intensity	uint16 / count	Uncalibrated amplitude; saturation flagged
angle / line_id	float32 rad / uint32	Water-side ray angle / line index
quality/status/pad	3 x uint16	Quality, health flags, reserved zero pad

Frame: +Z forward along optical axis; +X starboard; +Y down.
Packed point record: 40 bytes; 20k records/s -> 0.8 MB/s payload.
Add packet framing, sequence, CRC and calibration ID at transport layer.
No-return records carry validity flag; invalid coordinates use NaN.
Health stream: case/water temp, humidity, input V/I, interlock state.

04 ACCEPTANCE PLAN / UNRESOLVED ENGINEERING

V1 OPTICS / Ray trace through sapphire and seawater; verify full scan clearance, APD etendue, FOV overlap and internal scatter.

V2 RANGE / Tank target at 1, 5, 10, 15 m; record reflectance, attenuation, ambient light, water T/S and n_group model. >=1,000 valid returns/position; report bias, sigma and dropout.

V3 PRESSURE / External-pressure FEA, windows, glands and preload; establish proof protocol and fatigue life before hydrotest.

V4 THERMAL / Test 35 W continuous and 60 W peak duty in still water; verify source drift, APD noise and maximum case temperature.

V5 INTEGRATION / Timestamp alignment, EMC, leakage, corrosion, connector service life and hardware fail-off fault injection.

V6 SAFETY / Determine laser class for every accessible condition; release interlock design only after documented assessment.

REFERENCE BASIS: [R1] Non-coaxial underwater lidar research (2023) [R2] NIST speed of light. Targets above are original concept assumptions.

CONCEPT ONLY / NOT RELEASED FOR MANUFACTURE

ALL DIMENSIONS mm. NUMERIC DIMENSIONS GOVERN. DO NOT SCALE PDF.

Assumed geometry; pressure, optics, thermal performance and laser safety require validation.

DRAWING NRD-532-003
REV A / 2026-10-21
STATUS: PRELIMINARY

FORMAT A2
VIEW SCALE: AS NOTED
UNITS: mm / SI

SHEET 3 OF 3
ENGINEERING CONCEPT
NO TEST DATA